

Oppgave 1 La u være heavisidefunksjonen

$$u(t) = \begin{cases} 0 & \text{for } t < 0 \\ 1 & \text{for } t \geq 0 \end{cases}$$

a) Vis at

$$\mathcal{L}(u(t-a)) = \frac{e^{-as}}{s}, \quad \text{for } a \geq 0.$$

b) Løs initialverdiproblemet

$$y''(t) + y(t) = u(t-1) \quad y(0) = y'(0) = 0$$

og skisser løsningen.

$$a) \mathcal{L}(u(t-a)) = \int_a^\infty u(t-a) e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_a^\infty = \frac{e^{-sa}}{s}$$

$$b) y''(t) + y(t) = u(t-1), \quad y(0) = y'(0) = 0$$

Laplace transformerer IVP

$$s^2 F(s) + sF(0) + F'(0) + F(s) = \frac{e^{-s}}{s}$$

$$F(s)(s^2 + 1) = \frac{e^{-s}}{s}$$

$$F(s) = \frac{1}{(s^2 + 1)s} = \frac{As+B}{s^2+1} + \frac{C}{s} \Rightarrow 1 = As^2 + Bs + Cs^2 + C \Rightarrow \begin{matrix} A+C=0 \\ B=0 \\ C=1 \end{matrix} \Rightarrow A=-1, B=0, C=1$$

$$= e^{-s} \left(\frac{-s}{s^2+1} + \frac{1}{s} \right)$$

$$F(s) = e^{-s} F(s) \Leftrightarrow f(t) = u(t-1) f(t-1)$$

$$= u(t-1)(1 - \cos(t-1))$$

Oppgave 2 Denne teller som totalt en deloppgave.

a) Finn fourierrekken til funksjonen

$$f(x) = \sin(3x) + \sin(x) + 1$$

på intervallet $[-\pi, \pi]$.

b) KUN 4N: Finn fouriertransformen til funksjonen $f(x) = 6x \exp(-5x^2)$.

$$a) f(x) = \sin(3x) + \sin(x) + 1, \quad x \in [-\pi, \pi]$$

$$f(x) \sim a_0 + \sum_{n=1}^{\infty} a_n \cos(nx) + b_n \sin(nx)$$

$$a_0 = 1, \quad a_n = 0, \quad b_n = \begin{cases} 1, & n=3 \vee n=1 \\ 0, & \text{ellers} \end{cases}$$

$$b) f(x) = 6x e^{-5x^2} = -\frac{6}{10} \frac{d}{dx} e^{-5x^2}$$

$$F(f'(x)) = i\omega F(f(x))$$

$$-\frac{6}{10} F\left(\frac{d}{dx} e^{-5x^2}\right) = -\frac{6}{10} i\omega F(e^{-5x^2}) = -\frac{6}{10} i\omega e^{-\frac{\omega^2}{20}}$$

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{2\pi} \left(\int_{-\pi}^{\pi} \sin(3x) dx + \int_{-\pi}^{\pi} \sin(x) dx + \int_{-\pi}^{\pi} 1 dx \right) = \frac{1}{\pi} \int_0^{\pi} dx = \frac{1}{\pi} [x]_0^{\pi} = \frac{\pi}{\pi} = 1$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \frac{1}{\pi} \left(\int_{-\pi}^{\pi} \sin(3x) \cos(nx) dx + \int_{-\pi}^{\pi} \sin(x) \cos(nx) dx + \int_{-\pi}^{\pi} \cos(nx) dx \right) = \frac{2}{\pi} \int_0^{\pi} \cos(nx) dx = \frac{2}{\pi n} [\sin(nx)]_0^{\pi} = 0, \quad n \in \mathbb{Z}$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \frac{1}{\pi} \left(\int_{-\pi}^{\pi} \sin(3x) \sin(nx) dx + \int_{-\pi}^{\pi} \sin(x) \sin(nx) dx + \int_{-\pi}^{\pi} \sin(nx) dx \right) = \frac{2}{\pi} \left(\int_0^{\pi} \sin(3x) \sin(nx) dx + \int_0^{\pi} \sin(x) \sin(nx) dx \right) = \frac{2}{\pi} \left(\int_0^{\pi} \frac{\cos(x(3-n)) - \cos(x(3+n))}{2} dx + \int_0^{\pi} \frac{\cos(x(1-n)) - \cos(x(1+n))}{2} dx \right) = \frac{1}{\pi} \left(\left[\frac{\sin(x(3-n))}{3-n} - \frac{\sin(x(3+n))}{3+n} \right]_0^{\pi} + \left[\frac{\sin(x(1-n))}{1-n} - \frac{\sin(x(1+n))}{1+n} \right]_0^{\pi} \right)$$

Oppgave 3

Finn løsningen til bølgeklningen

$$u_{tt} = u_{xx},$$

for $0 \leq x \leq \pi$ og $t \geq 0$, med randkrav

$$u(0, t) = u(\pi, t) = 0$$

og initialkrav

$$u(x, 0) = \sin x, \quad u_t(x, 0) = 0.$$

$$u_{tt} = u_{xx}$$

Bruker d'Alembert's

$$u(x, t) = \frac{1}{2} (f(x+ct) + f(x-ct)) + \frac{1}{2c} \int_{x-ct}^{x+ct} g(\tilde{x}) d\tilde{x}$$

$$u(x, t) = \frac{1}{2} (\sin(x+t) + \sin(x-t)) = \sin(x) \cos(t)$$

Oppgave 4 Utled løsningsformelen

$$u(x, t) = \frac{1}{2c\sqrt{\pi t}} \int_{-\infty}^{\infty} f(v) e^{\frac{-(x-v)^2}{4c^2 t}} dv, \quad t > 0,$$

for varmelikningen

$$u_t = c^2 u_{xx},$$

på hele x -aksen med initialkrav

$$u(x, 0) = f(x).$$

Wtaf.

Oppgave 5 Finn polynomet av grad 3 som interpolerer $f(x) = e^x$ i punktene $x = 0, x = 1, x = 2$ og $x = 3$.

x	0	1	2	3
y	1	e	e ²	e ³

$$L_0 = \left(\frac{x-x_1}{x_0-x_1} \right) \left(\frac{x-x_2}{x_0-x_2} \right) \left(\frac{x-x_3}{x_0-x_3} \right) = \left(\frac{x-1}{0-1} \right) \left(\frac{x-2}{0-2} \right) \left(\frac{x-3}{0-3} \right) = -\frac{x^3}{6} + x^2 - \frac{3x}{2} + \frac{2}{3}$$

$$L_1 = \left(\frac{x-x_0}{x_1-x_0} \right) \left(\frac{x-x_2}{x_1-x_2} \right) \left(\frac{x-x_3}{x_1-x_3} \right) = \left(\frac{x-0}{1-0} \right) \left(\frac{x-2}{1-2} \right) \left(\frac{x-3}{1-3} \right) = \frac{x^3}{2} - \frac{5x^2}{2} + 3x$$

$$L_2 = \left(\frac{x-x_0}{x_2-x_0} \right) \left(\frac{x-x_1}{x_2-x_1} \right) \left(\frac{x-x_3}{x_2-x_3} \right) = \left(\frac{x-0}{2-0} \right) \left(\frac{x-1}{2-1} \right) \left(\frac{x-3}{2-3} \right) = -\frac{x^3}{2} + 2x^2 - \frac{3x}{2}$$

$$L_3 = \left(\frac{x-x_0}{x_3-x_0} \right) \left(\frac{x-x_1}{x_3-x_1} \right) \left(\frac{x-x_2}{x_3-x_2} \right) = \left(\frac{x-0}{3-0} \right) \left(\frac{x-1}{3-1} \right) \left(\frac{x-2}{3-2} \right) = \frac{x^3}{6} - \frac{x^2}{2} + \frac{x}{3}$$

$$L(x) = \sum L_j y_j = \text{Lagt utr.} \Rightarrow \text{rett}$$

Oppgave 6 Vis at

$$\frac{f(x+h) - 2f(x) + f(x-h))}{h^2}$$

er en andre ordens approksimasjon til $f''(x)$. *Hint:* Bruk Taylor-rekker. Du kan forutsette at f er tilstrekkelig glatt.

$$f(x+h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} h^n = f(x) + f'(x)h + \frac{f''(x)}{2} h^2$$

$$f(x-h) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!} (-h)^n = f(x) - f'(x)h + \frac{f''(x)}{2} h^2$$

$$f(x+h) + f(x-h) = f(x) + \cancel{f'(x)h} + \frac{f''(x)}{2} h^2 + (f(x) - \cancel{f'(x)h} + \frac{f''(x)}{2} h^2)$$

$$f''(x) = \frac{f(x+h) + f(x-h) - 2f(x)}{h^2}$$